
Probability and Statistics for Engineers

Reasoning under uncertainty

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A note on names

Los Angeles' longer Spanish name

El Pueblo de Nuestra Señora la Reina de los Ángeles de Porciúncula

You probably call it

L.A.

My full name

José Antonio Aguirre Pérez de León

You can call me

Antonio or Tony

PhD candidate in Statistics, UC Santa Cruz

From Mexico City. Background in applied mathematics and economic theory.

My research

Forecasting	I develop deep learning and Bayesian models for data that change over time.
Applications	Correcting environmental and climate forecasts and quantifying uncertainty.
Software	I develop statistical software in R for fitting and evaluating models.

The San Lorenzo River

What is the probability that river flow exceeds a critical level?

Probability questions in computing

Google

Response times

Google's SRE book explains why an average response time can hide slow requests.

How often will a user wait much longer than usual?

Stripe

Payment
screening

Stripe Radar evaluates payment information to assess the risk of fraud.

If a payment is flagged, how likely is it to be fraudulent?

Sources: Google, *Site Reliability Engineering*, Ch. 6; Stripe, *Transaction risk prevention*.

Same average, different waiting times

An online store tests two versions of its product search.

How long does a customer wait for the results?

Waiting time in seconds

	Search 1	Search 2	Search 3	Search 4	Average
Version A	1.0	1.0	1.0	1.0	1.0
Version B	0.2	0.2	0.2	3.4	1.0

The store wants results within 2 seconds

Which version meets that goal more often in these four tests?

Outcomes, sample spaces, and events

Outcome ω

One possible result of the experiment.

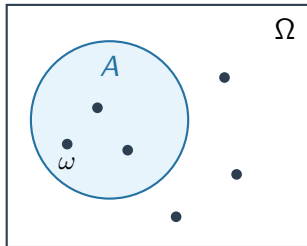
Sample space Ω

The set of all possible outcomes.

Event $A \subseteq \Omega$

A subset of Ω . It occurs when the observed outcome belongs to it.

Today Ω is finite and nonempty. Every subset is an event.



A finite probability model

Assign a probability $q_\omega = P(\{\omega\})$ to each outcome $\omega \in \Omega$.

Conditions on the outcome probabilities

$$q_\omega \geq 0 \quad \text{for every } \omega \in \Omega, \quad \sum_{\omega \in \Omega} q_\omega = 1.$$

For an event A , add the probabilities of its outcomes:

$$P(A) = \sum_{\omega \in A} q_\omega.$$

Equally likely outcomes are a modeling assumption.

Two items, four locations

A simple allocation model

Place two distinct items into locations 1, 2, 3, 4.

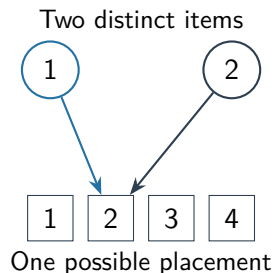
Uniform choices: each location has probability $1/4$ for each item.

Independence: the first choice does not change the probabilities for the second.

Thus each ordered pair (i, j) has probability $(1/4)(1/4) = 1/16$.

Collision: both items receive the same location.

Motivated by hashing. The uniform, independent allocation is an idealized model.



The sample space as a grid

		Second item's location			
		1	2	3	4
First item's location	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)

$$\Omega = \{(i, j) : i, j \in \{1, 2, 3, 4\}\}.$$

There are $4 \times 4 = 16$ possible outcomes.

The order records *which item* receives each location.

Which outcomes correspond to a collision?

$$A = \{ \quad \quad \quad \}$$

The collision probability

		Second item's location			
		1	2	3	4
First item's location	1	(1, 1)	(1, 2)	(1, 3)	(1, 4)
	2	(2, 1)	(2, 2)	(2, 3)	(2, 4)
	3	(3, 1)	(3, 2)	(3, 3)	(3, 4)
	4	(4, 1)	(4, 2)	(4, 3)	(4, 4)

The diagonal is the collision event:

$$A = \{(1, 1), (2, 2), (3, 3), (4, 4)\}.$$

Each cell has probability $1/16$.

$$P(A) = \sum_{\omega \in A} q_{\omega}$$

For a finite space with equally likely outcomes, $P(A) = |A|/|\Omega|$.

The complement rule

$A^c = \Omega \setminus A$ contains the outcomes outside A .

A and A^c are disjoint, and together they contain every outcome.

$$1 = P(\Omega) =$$

For every event in our finite model

$$P(A^c) = 1 - P(A).$$

Here, $P(\text{no collision}) =$.

Practice: a collision in location 1 or 2

Use the same uniform, independent allocation model.

- 1 Write the event B that the items collide in location 1 or location 2.
 - 2 Find $P(B)$ and explain your reasoning.
 - 3 How would the method change if the outcome probabilities were unequal?
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Ungraded practice. Compare your reasoning with a neighbor.

Questions we will learn to answer

Question

Probability tools

What can happen?

Events, counting, probability models

What does new information change?

Conditioning and Bayes' theorem

How do numerical outcomes vary?

Distributions, expectation, variance

What do repeated observations tell us?

Limits, estimation, uncertainty

How do random systems evolve?

Poisson processes, Markov chains

The same habit throughout: state the model, reason from it, interpret the result.

Course references

Morris H. DeGroot and Mark J. Schervish

Probability and Statistics, fourth edition.

Mor Harchol-Balter

Introduction to Probability for Computing, 2024.

Canvas lists the assigned sections, exercises, and reading resources.

Today's background: DeGroot–Schervish, Sections 1.3–1.6; Harchol-Balter, Sections 2.1–2.2. The allocation example is motivated by Section 20.2.

Additional teaching reference: Sheng Jiang, *STAT 131 Lecture Notes*, July 18, 2024, pp. 3–4.

How we will work

Lectures	Definitions, worked examples, and time to reason together.
Discussions	Optional TA-led recap, examples, and practice. No attendance requirement, submission, or grade.
Independent work	Homework and a short weekly quiz.

Please do not use Canvas messages

Use jaguir26@ucsc.edu for individual questions and Ed Discussion for course questions.

TA assignments, discussion times, and office hours: TBD. Consult the syllabus for full policies.

Before Monday

In Canvas, open Start Here.

- ➊ Read the syllabus and check how the course is organized.
- ➋ Begin the ungraded mathematics review. Mark the topics where you would like a refresher.
- ➌ Bring your questions. On Monday, we begin with events and counting.

Today's essentials

Identify the possible outcomes and the event of interest.

State the probability model and its assumptions.

Add the probabilities of the outcomes in the event.